

Ali Nasr

AI Systems Architect | R&D Engineer

ainlamyae@gmail.com | [Website](#) | [GitHub](#) | [LinkedIn](#) | [Google Scholar](#)

About

AI Systems Architect and R&D Engineer with extensive expertise in the design and MBSE of autonomous platforms, from agentic or intelligent decision-making to real-time robotic control.

Core Engineering Expertise

- Requirements Engineering, System & Component Architecture Design
- Prompt Engineering, Multi-Agent Orchestration, Reliability Pipelines, Human-in-the-Loop Validation
- Modeling, Identification, Training, Validation, and Simulation
- Software-in-the-loop Testing, Real-time Control Development, Signal Processing, Runtime Monitoring
- Embedded Software, Communication Systems, Sensor & Actuator Integration
- Calibration, Prototyping, and Optimization

Systems Experience

- **Generative AI:** Text Generation, Code Generation (5G Scripts & Test Cases), NL Interpretation, RAG, LLM-as-a-Judge
- **Intelligent Systems:** Autonomous Driving (Apollo Auto, End-to-End Interfuser)
- **Robotics:** Industrial, Serial & Parallel (Cable, Stewart), Humanoid, Biped, UGV, UAV (Quadcopter), Exoskeleton (Upper & Lower Limb)
- **Biological Systems:** Musculoskeletal Modeling, sEMG Analysis
- **Physical Systems:** Electrical, Hydraulic, Pneumatic, Mechanical

Skills

- **Generative AI:** LangChain, Ollama
- **Machine Learning:** ANN, RNN, CNN, RCNN, LSTM, GRU, SVM
- **Programming:** Python (NumPy, SciPy, Matplotlib, Pandas, scikit-learn), MATLAB, C++
- **System Modeling & Simulation:** MATLAB (Simulink, Simscape Multibody, Deep Learning Toolbox, Neural Networks, GA, PSO), MapleSim, LabVIEW
- **Embedded Systems:** Raspberry Pi, NVIDIA Jetson, Arduino, AVR Microcontrollers, PLCs (Siemens, Delta, Vigor)
- **Control Methods:** PID, LQR, LQG, CTM, MPC, NMPC, Feedback Linearization, Impedance, Adaptive, Sliding Mode, Fuzzy Logic, ANFIS
- **CAD & Mechanical Design:** CATIA, SolidWorks
- **Hands-on Skills:** Mechanical Fabrication, Soldering, 3D Printing

Experience

Senior R&D Engineer (GenAI System, Assurance Architect, and Design) 2023-11-01 – Present

Huawei Technologies Canada — Waterloo, Ontario, Canada

Permanent Full-time

Trustworthy Technology and Engineering Laboratory, Waterloo Complex Intelligent Systems Laboratory
LLMGuard Wrapper: A Runtime Assurance Framework for Reliable LLM Systems

- Designed and delivered a runtime monitoring and assurance architecture (GuardRailClient) for LLM-

based multi-agent systems, enabling reliable and production-ready AI behavior through integrated monitoring, validation, and correction

- Introduced a novel confidence-driven methodology that leverages token-level LogProbs aggregated into JSON-value scores to detect uncertainty and autonomously identify erroneous outputs without labeled data
- Developed a modular, scalable framework (wrapper-based integration) that seamlessly plugs into existing LLM pipelines with minimal code changes, supporting both synchronous and asynchronous execution
- Implemented an adaptive validation and recovery pipeline using LLM-as-a-Judge, multi-strategy correction (prompting, model switching, self-consistency), and optional human-in-the-loop escalation for high-risk cases
- Engineered an efficient field-level correction mechanism that selectively fixes low-confidence outputs, reducing recomputation, lowering latency, and improving system stability in real-world deployments
- Demonstrated measurable reliability improvements in a script generation system, successfully detecting ambiguous/erroneous outputs and enabling automated correction under real-world and adversarial scenarios

Uncertainty in LLMs: Theory, Methods, and Practical Reliability Frameworks

- Conducted a technical review and synthesis of uncertainty in LLMs, covering theory, state-of-the-art methods, and practical applications for improving reliability in real-world AI systems
- Analyzed and formalized key uncertainty concepts (aleatoric, epistemic, semantic, entropy-based metrics), translating research insights into actionable frameworks for inference-time risk assessment
- Evaluated and compared multiple uncertainty estimation techniques (token probability, entropy, sampling, self-check, and ensemble methods), identifying strengths, limitations, and failure modes such as "confidently wrong" predictions
- Designed and executed experiments on hallucination detection using uncertainty features, including entropy, token probabilities, and Dirichlet-based metrics, demonstrating effective model-agnostic detection without ground truth
- Developed and validated a multi-feature classification approach for hallucination detection, showing improved performance over single-metric methods and highlighting the importance of combining uncertainty signals
- Derived system-level insights for production AI systems, including the impact of temperature on uncertainty, limitations of entropy-based metrics, and the need for multi-step uncertainty tracking in agent-based workflows

Explainable & Reliable System Architecture for LLM-based Script Generator

- Architected a multi-stage LLM reliability pipeline with confidence estimation, output gating, and policy-driven decision enforcement
- Implemented probability-based and token-level uncertainty scoring to dynamically control acceptance, remediation, or escalation of LLM outputs
- Applied voting and consensus mechanisms across multiple LLM outputs to improve consistency and select highest-agreement results
- Engineered an assurance loop with temperature adjustment, prompt perturbation, and model switching for iterative refinement
- Implemented a Decision Maker using a Markov Decision Process (MDP) for policy enforcement under system constraints
- Enabled fail-safe escalation paths including re-execution with alternative models, LLM-as-Judge evaluation, and human-in-the-loop review
- Designed an explainable JSON-based output schema ensuring traceability of generated variables and intermediate decisions
- Proposed metamorphic prompt perturbation via semantic-preserving sentence swapping (Chinese inputs) to test robustness and consistency

Multi-Agent Foundation Model-based AI for Autonomous Software Testing Platform

- Designed a multi-agent system architecture leveraging foundation model-based AI (RAG-based LLMs) for a fully autonomous software testing platform
- Developed run-time accuracy and consistency monitoring mechanisms for collaborative AI agents
- Collaborated on embodied AI concepts for general-purpose intelligent assistance systems

2D Bird's-Eye View Risk Monitoring for Autonomous Driving

- Developed a CNN-based risk assessment module that processes 2D bird's-eye views of the vehicle's surroundings to predict the future driving risk
- Implemented risk detection using a 2D Time-to-Collision (TTC) algorithm for perceived object and obstructed phantom object
- Coded and integrated C++ modules into Baidu Apollo Auto ADS, enabling real-time risk evaluation and decision-making
- Verified and tested the system under diverse conditions, analyzing accident scenarios and validating robustness of the risk detection algorithms
- Animated 2D bird's-eye views and reconstructed accident cases from China for kinematic analysis and validation

Game Theory Orchestration of Multi-Agent LLM-based Reinforcement Learning

- Proposed and scoped a game-theoretic orchestration framework for multi-agent LLM-based reinforcement learning systems

Uncertainty Monitoring System for LLM-Based Generative Software

- Implemented the core uncertainty computation for the LLM outputs, forming the foundation for the monitoring system
- Developed the decision-making logic to act on uncertainty measures and partially contributed to coding the overall system
- Assisted in project planning and integration, supporting the application of uncertainty metrics to JSON-based outputs

Run-time Behavioural Validation of Self-adaptive AI-enabled Systems

- Led and coordinated the uOttawa engineering team, aligning technical efforts and ensuring milestone delivery
- Extended metamorphic testing methodologies to LLM-based software and defined practical application strategies
- Collected and formalized metamorphic relations for LLM applications to detect subtle undesirable behaviors beyond explicit failures
- Supervised development of a search-based test generation approach for autonomous driving scenarios
- Oversaw case studies implemented in CARLA simulator on the Interfuser autonomous driving system
- Directed automated offline system-level testing combining Metamorphic Testing with Evolutionary Algorithms

LLM-based System Reliability

- Guided implementation of probability extraction from LLMs within LangChain

C++ Real-Time Processing Engine for Monitoring Software

- Developed prototype of C++ modules for processing as part of the monitoring software project
- Conducted testing and validation of the C++ components to ensure reliability before integration
- Delivered ready-to-use modules to assist in integration into the larger framework

Strategic Insights and Innovations from Global Technology Conferences

- 14th International Conference on Learning Representations (ICLR), Rio de Janeiro, Brazil, Apr 2026
- 39th Annual Conference on Neural Information Processing Systems (NeurIPS), Mexico City, Mexico, Nov–Dec 2025
- 34th International Joint Conference on Artificial Intelligence (IJCAI), Montreal, Canada, Aug 2025
- 38th Annual Conference on Neural Information Processing Systems (NeurIPS), Vancouver, Canada, Dec 2024
- OpenMind Research Institute Banff Retreat, Banff, Canada, Oct 2024
- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, USA, Jun 2024
- Systems Engineering Workshop for Complex Intelligent Systems, Hong Kong Science Park, Hong Kong, May 2024
- Symposium on Innovative Robotics Technologies and Applications, Hong Kong Science Park, Hong Kong, May 2024
- Strategy and Technology Workshop by Huawei R&D Teams, Shenzhen and Dongguan, China, May 2024
- 18th Annual IEEE International Systems Conference (SysCon), Montreal, Canada, Apr 2024
- 38th Annual AAAI Conference on Artificial Intelligence (AAAI), Vancouver, Canada, Feb 2024

Project Manager & Technical Interviewer Responsibilities

- Managed project workflows using Jira, tracked progress, led status updates, and prepared detailed project progress reports
- Identified team skill gaps and collaborated with leadership to define role requirements, and draft job descriptions
- Screened 1,000+ resumes, and shortlisted high-potential candidates based on technical qualifications
- Conducted 50+ remote technical interviews, evaluating problem-solving and programming ability

Senior R&D Engineer (AI Assurance in Autonomous Driving) 2023-06-01 – 2023-11-30
Huawei Technologies Canada — Waterloo, Ontario, Canada

Contract Full-time

Trustworthy Technology and Engineering Laboratory, Waterloo Complex Intelligent Systems Laboratory

Run-time Assurance (AI Consistency and Accuracy) System for Autonomous Driving Systems

- Researched failure measurement and monitoring of learning-enabled control systems across Perception, Prediction, and Planning modules
- Performed systems-theoretic process analysis and defined risk boundaries for autonomous driving systems
- Formulated and implemented C++ algorithms for monitoring, risk assessment, and secure disengagement protocols
- Designed a comprehensive run-time assurance architecture for AI consistency and accuracy validation
- Evaluated advanced autonomous driving modules by integrating Baidu Apollo Auto with CARLA Simulator using C++ and Python
- Designed pre-crash, adverse weather, and empirical simulation scenarios to validate robustness and safety

Run-time Assurance (Vehicle Safety) System for Autonomous Driving Systems

- Designed, developed, and tested vehicle recovery software for safety violations in autonomous driving systems
- Implemented proactive risk prediction and adaptive threshold tuning mechanisms
- Extracted requirements from ISO 26262, ISO 25058, and AVSC 6202103 to design compliant safety-oriented software architecture
- Developed AI system testing strategies including white-box and black-box evaluation for behavioral and activation coverage analysis

- Prototyped system components in Python and C++ and collaborated with simulation and production teams for validation

Postdoctoral Research Associate (Autonomous Control)

2023-01-01 – 2024-03-31

University of Waterloo — Waterloo, Ontario, Canada

Contract Full-time

Department of Systems Design Engineering *Real-time Embedded Controller within Different Sensors and Intelligence Control*

- Successfully designed and implemented platform real-time communication and control software on Raspberry Pi 4B (Linux environment) to support advanced application features, resulting in a 20% increase in user engagement and a 30% reduction in software crashes
- Developed machine learning-based real-time control unit with Python (multi threaded and object-oriented), resulting in improved system performance and accuracy
- Proficient in developing mid-level programs with Bluetooth, CAN-bus, and SPI connections to achieve seamless communication between hardware and software
- Conducted rigorous feature validation using hardware-in-the-loop (HIL) testing, ensuring the robustness and reliability of the system

Electrical Energy Regenerative Shoulder Exoskeletons

- Implementation of energy-regenerating actuators in robotic exoskeletons to enhance efficiency and extend battery life by converting a portion of energy typically lost during negative mechanical work into electrical energy.
- Evaluated the electrical energy regeneration capabilities of a shoulder wearable exoskeleton.
- Motor parameters were determined through experiments using a joint dynamometer system.
- Developed an electro-mechanical model for human-exoskeleton interaction focusing on energy regeneration.
- Conducted task-based experiments to measure energy regeneration during real-life usage of the exoskeleton.
- Results compared with practical usage scenarios to assess feasibility; electricity generated was relatively low due to motor characteristics, but indicated potential for energy regeneration with certain limitations.

Human3.6M+: Generated Human Motion Data to Support 3D Pose, Motion, Kinetic, and Muscle Estimation

- Developed marker-less motion capture technology for 3D human pose estimation to overcome usability constraints of traditional marker-based systems.
- Synthesized dynamics and muscle properties from existing motion capture datasets to generate training and validation data for deep learning models.
- Reconstructed motions using kinematics, dynamics, and muscle modeling while ensuring adherence to International Society of Biomechanics standards.
- Integrated musculoskeletal model with customizable anthropometric features (height, mass, athleticism level, handedness, skin temperature) to enhance realism.
- Applied methodology to Human3.6M dataset, demonstrating kinematic similarity to ground truth markers and dynamic plausibility compared to experimental data.

Postdoctoral Responsibilities

- Organized a remote international workshop with Prof. Daniel J. Rixen's group from Technical University of Munich (TUM)
- Coordinated presentations on topics ranging from humanoid robot demonstrations to advanced research in biomechanics and control systems

Myography-based Controlled 1-DoF Hand Prosthesis

- Designed an electrically controlled single degree-of-freedom (1-DoF) forearm-hand prosthesis to aid pediatric amputees after Mine Blast Injuries in Ukraine.
- Achieved precise thumb controllability through electrical component development, control system refinement, mechanical design, and final assembly.
- Prototype serves as an active pediatric prosthesis using EMG-based control rather than muscle strength, despite being limited to a single finger.
- Future improvements planned: refine 3D model, optimize electronic housing, enhance EMG control precision, and incorporate motors for full-finger control.
- Prototype is a pivotal tool for initiating prosthesis control learning in rehabilitation.

MyoBuzz: Real-time Muscle Activation Monitoring in Dynamometry Experiments

- Developed MyoBuzz, a limb-worn EMG device providing real-time feedback on muscle activation for dynamometry experiments.
- Limb-specific band with strategically positioned electrodes, visual and auditory alerts, customizable activation threshold, and adjustable sensitivity.
- Electrodes are attachable via clips; strap ensures comfort; rechargeable 3.7V battery supports one-hour operation and two-hour recharge.
- Facilitates real-time monitoring and mitigation of inadvertent muscle activation during joint torque evaluations.

Human Motion Animation Using 3Ds Max for 3D Pose Estimation Database

- Guided in setting up 3Ds Max environment and workflow for creating human motion animations for synthetic dataset generation
- Provided joint position data and guidance to support annotation for training the 3D pose estimation machine learning model

Graduate Teaching Assistant

2019-01-01 – 2023-04-30

University of Waterloo — Waterloo, Ontario, Canada

Contract Part-time

Department of Systems Design Engineering *Teaching Responsibilities*

- Delivered tutorials and undergraduate instructional support
- Explained complex engineering concepts during laboratory sessions, office hours, and via email communication
- Evaluated assignments, projects, and examinations to assess student learning outcomes

Course List **Graduate Research Assistant (BioMechatronic Systems)** 2018-09-01 – 2022-12-31

University of Waterloo — Waterloo, Ontario, Canada

Contract Full-time

Motion Research Group *Design, Dynamics, and Control of Active-Passive Exoskeletons Robotic*

- Developed and verified adjustable biomechanical 20-DoF model using MATLAB and MapleSim
- Designed, optimized, and developed actuator of wearable robot in accordance with ISB standards
- Conducted research and developed predictive models using ANN, RNN, CNN, and RCNN to estimate biomechanical (kinematic and kinetic) variables from noisy electromyography sensor data
- Developed novel assist-as-needed control regulations for wearable robots using model-based predictive control and fuzzy logic techniques
- Implemented torque and motion real-time control of active wearable robots with sEMG sensors and CAN-bus connection, leading to a 15% improvement in accuracy and a 25% increase in robot speed
- Designed and implemented safety measures for autonomous systems to ensure compliance with human testing, industry regulations, and standards, resulting in successful attainment of safety approval and ethics certificate

Active-passive Shoulder Exoskeleton

- Developed active-passive shoulder exoskeleton integrating passive mechanisms and powered actuators to optimize human movement assistance.
- Established multibody kinematics and dynamics model for upper body, implemented optimal controller for reducing muscular activation torque.
- Optimized exoskeleton design to enhance user comfort and reduce fatigue while maintaining functional assistance.
- Results demonstrate substantial reduction in fatigue and functional improvement over fully passive or fully active systems.

MuscleNET: Biomechatronic Electromyography Signal Processing

- Introduced MuscleNET, a set of ML models (ANN, RNN, CNN, RCNN) mapping sEMG signals to joint angle, velocity, acceleration, torque, and activation torque.
- RobustMuscleNET RNN predicts control-oriented signals in real-time with 92% regression accuracy for joint angle and 86.5% for assistive torque.
- InverseMuscleNET resolves redundancy in inverse muscle models using RNNs, achieving 88–91% normalized regression between estimated and experimental muscle activation.
- Enables fast, direct estimation for real-time applications in rehabilitation and sports evaluation without invasive electrodes.

Scalable Human Musculoskeletal Model

- Developed a MapleSim biomechanics-based musculoskeletal (MSK) model covering upper and lower body movements for biomechanical analysis.
- Estimated biomechanical joint torque using muscle torque generators with active and passive components.
- Created adjustable multibody kinematic model accommodating diverse anthropometric measurements (sex, age, body mass, height, dominant side, physical activity, skin temperature) with 40 DoF.
- Validated model equations through simulations of joint range of motion and torque, demonstrating robust agreement with existing research findings.

Conferences & Academic Activities

- 18th International Symposium on Computer Methods in Biomechanics and Biomedical Engineering, Paris, France, May 2023
- Waterloo Robotics Day, Waterloo, Canada, March 2023
- 6th International Conference on Multibody System Dynamics, New Delhi, India, October 2022
- CRE-MSD Conference on Manual Handling, Mississauga, Canada, October 2022
- 5th North American Congress on Biomechanics, Ottawa, Canada, August 2022
- SYDE Graduate Symposium, Waterloo, Canada, June 2022
- 45th American Society of Biomechanics Annual Conference, Virtual, August 2021
- 45th Mechanisms and Robotics Conference, Virtual, August 2021
- 17th International Conference on Multibody Systems, Nonlinear Dynamics, and Control, Virtual, August 2021
- 7th International Conference of Control, Dynamic Systems, and Robotics, Virtual, November 2020
- 16th Annual Ontario Biomechanics Conference, Nottawasaga, Canada, March 2019

R&D Mid-Level Engineer (Robotics)

2016-01-01 – 2018-08-31

Techno Niro — Dorcheh, Esfahan, Iran

Contract Full-time Spatial 6 DoF 8 Cable-Driven Parallel Robot

- Conducted comprehensive inverse kinematic and dynamic modeling and analysis for the Spatial 6 DoF 8 Cable-Driven Parallel Robot.
- Validated the developed models using the SimMechanics Toolbox.
- Designed an optimal cable structure attachment for the robot.

- Simulated PD Partial, LQR, Robust, and Adaptive Control in the presence of noise and disturbance.

3 DoF Wrist Rehabilitation Robot

- Designed conceptual and mechanical parts for the 3 DoF Wrist Rehabilitation Robot and conducted kinematic and dynamic modeling.
- Implemented modified impedance control for enhanced human-robot interaction.
- Prototyped the wrist rehabilitation robot (Jan 2016 – Jan 2017), integrating modeling, mechanical design, and control for testing.

Graduate Marker

2015-09-01 – 2015-11-30

K. N. Toosi University of Technology — Tehran, Tehran, Iran

Remote

Faculty of Mechanical Engineering *Teaching Responsibilities*

- Assessed student learning through evaluating assignments, projects, and exams

Course List **Graduate Research Assistant (System Control)**

2014-09-01 – 2015-12-31

K. N. Toosi University of Technology — Tehran, Tehran, Iran

Contract Part-time

Advanced Robotics & Automated Systems (ARAS) Laboratory *Multiple Impedance Control of Hybrid Parallel-Serial Manipulator for Object Manipulation*

- Designed and developed a MATLAB/Simulink model for an 8 cable-driven parallel robot prototype with a serial manipulator, and verified its performance
- Conducted a comprehensive analysis of stiffness, sensitivity, singularity, and workspace volume to optimize the structure of a constraint and suspended parallel robot, using Genetic Algorithm (GA)
- Simulated and evaluated multiple impedance control algorithms, including serial and parallel, for object manipulation and human-robot force/moment interaction
- Prototyped the 8 cable constraint parallel robot, integrated image processing (vision) and IMU modules for precise position and orientation feedback, proposed an algorithm to address sensor shortage, and successfully implemented real-time motion control

Optimal Redundancy Resolution Scheme in a Cable-Driven Parallel Robot

- Developed an enhanced kinematic modeling of pulleys and cables for more accurate representation of cable-driven parallel robots
- Improved tension distribution analysis by integrating precise cable attachment modeling into the system framework
- Implemented a basic optimization approach using `fmincon` for estimating actuator force distribution

Sensory Feedback and Experimental Evaluation for Cable Robots with Uncertain Jacobian

- Led the mathematical modeling, algorithm coding, and manufacturing of the prototype of the cable-driven parallel robot
- Integrated the experimental and theoretical foundations on the robot's design and optimization approach
- Assisted the team with hardware, software, and modeling efforts

Visual Servoing in a Cable Robot Using Microsoft Kinect V2 Sensor

- Developed the kinematic model and control code for the parallel cable robot
- Supported integration between the robot and the visual tracking system
- Guided the use and modification of the Microsoft Kinect sensor for experiments

Digital Image Processing of Ball-and-Plate System Using Artificial Neural Network

- Integrated camera with MATLAB for real-time ball position tracking using classical image processing

techniques

- Contributed to development of a vision-based closed-loop control system
- Improved stability and responsiveness of ball positioning on a two-axis tilting platform

R&D Junior Engineer (Robotics)

2013-09-01 – 2015-12-31

Techno Niro — Dorcheh, Esfahan, Iran

Contract Part-time 6 DoF Industrial Serial Manipulator Modeling & Control

- Conducted comprehensive kinematic and dynamic modeling and analysis for the 6 DoF Industrial Serial Manipulator, assessing workspace, dexterity, and performance
- Utilized CATIA software for the design of mechanical parts, performed finite element analysis, and supervised the manufacturing process using CNC machines
- Prepared the hardware-in-loop of the control system using LabVIEW and MATLAB, and conducted motor parameters identification
- Implemented PID control, adjusted control coefficients for different maneuvers, and planned trajectory using ARM-controller programming

R&D Intern (Automation)

2013-06-01 – 2013-08-31

Kian Control Apadana Co. (Festo Authorized Distributor) — Isfahan, Esfahan, Iran

Internship Automation and Mechatronic System Development

- Developed the mechatronic scheme for a steel plant using ePLAN software, including designing the sensors, actuators, and wiring for the system
- Conducted extensive research on the hydraulic variable displacement pump action, exploring various modes of operation and optimizing their efficiency

Undergraduate Marker

2013-01-01 – 2013-05-31

Isfahan University of Technology — Isfahan, Esfahan, Iran

Hybrid Teaching Responsibilities

- Assessed student learning by marking assignments

Course List Undergraduate Research Assistant

2012-09-01 – 2013-08-31

Isfahan University of Technology — Isfahan, Esfahan, Iran

Contract Part-time

Department of Mechanical Engineering *Joint Space Energy Optimization of Tethered Space Robot for End-Effector Optimal Path*

- Conducted research on tethered satellite systems and tether-connected space manipulators for applications such as satellite repair and space debris removal.
- Modeled a tethered space manipulator system using a straight, rigid rod for the tether, restricting motion to the orbital plane with the platform following a circular orbit around Earth.
- Analyzed system dynamics with the end-effector tracing prescribed trajectories, monitoring tether tension to prevent compressive forces.
- Investigated feasible end-effector trajectories under torque restrictions, ensuring torques on tether attachment points were zero.
- Proposed a method to determine possible end-effector paths between two points while satisfying the zero-torque constraint on the tether.
- Developed and simulated a nonlinear control algorithm for vibration reduction in a redundant robot by adjusting cable length, validated through simulations and experiments.
- Performed comprehensive modeling of forward kinematics, workspace, and dynamics of a 4-DoF tethered space robot to evaluate performance and energy efficiency.

R&D Intern

2009-06-01 – 2013-08-31

Techno Niro — Dorcheh, Esfahan, Iran

Contract Part-time Plant Mechatronic Design, Maintenance, and Project Management

- Developed and supervised the implementation of a comprehensive mechatronic scheme for the plant, utilizing ePlan software and programming Siemens PLC for optimal performance
- Provided proactive maintenance and effective troubleshooting for mechanical systems and equipment
- Actively participated in project planning and execution, fostering collaboration to meet project deadlines
- Ensured adherence to quality standards, conducted regular inspections, and maintained accurate documentation of all mechanical systems
- Stayed abreast of industry trends and advancements, and pursued continuous learning through workshops and training

Education

Doctor of Philosophy (Ph.D.) in Systems Design Engineering 2018-09-01 – 2022-12-30
University of Waterloo — Waterloo, Ontario, Canada

- Major: Intelligent & Automated Systems; Biomechatronics Systems
- GPA: 88/100
- Thesis: [Design, Dynamics, and Control of Active-passive Upper-limb Exoskeleton Robots](#)

Master of Science (M.Sc.) in Mechatronics Engineering 2013-09-01 – 2015-12-30
K. N. Toosi University of Technology — Tehran, Tehran, Iran

- Major: Design Robotic and Mechatronic Systems
- GPA: 19.2/20
- Thesis: [Multiple Impedance Control of Hybrid Parallel-Serial Manipulator for Object Manipulation](#)

Bachelor of Science (B.Sc.) in Mechanical Engineering 2009-09-01 – 2013-08-30
Isfahan University of Technology — Isfahan, Esfahan, Iran

- Major: Mechatronics and Dynamic Systems
- GPA: 16.67/20
- Thesis: [Optimizing Joint Space Energy of Redundant Tethered Space Robots](#)

Publications

1. Yousefizadeh, Hossein, Gu, Shenghui, Briand, Lionel C., **Nasr, Ali**. *Constrained co-evolutionary metamorphic differential testing for autonomous systems with an interpretability approach*. Transactions on Software Engineering and Methodology (TOSEM) (2026).
2. Yousefizadeh, Hossein, Gu, Shenghui, Briand, Lionel C., **Nasr, Ali**. *Using cooperative co-evolutionary search to generate metamorphic test cases for autonomous driving systems*. (2026).
3. Yousefizadeh, Hossein, Gu, Shenghui, Briand, Lionel C., **Nasr, Ali**. *Using cooperative co-evolutionary search to generate metamorphic test cases for autonomous driving systems*. Transactions on Software Engineering (2025). [doi:10.1109/TSE.2025.3570897](https://doi.org/10.1109/TSE.2025.3570897)
4. **Nasr, Ali**, McPhee, John. *Harnessing biomechanical energy using shoulder exoskeleton: Electrical energy regeneration*. Journal of Mechanisms and Robotics (2025). [doi:10.1115/1.4068430](https://doi.org/10.1115/1.4068430)
5. **Nasr, Ali**, Inkol, Keaton A., McPhee, John. *Safety in wearable robotic exoskeletons: Design, control, and testing guidelines*. Journal of Mechanisms and Robotics (2025). [doi:10.1115/1.4066900](https://doi.org/10.1115/1.4066900)
6. **Nasr, Ali**, Risacher, Frederic. *Method and a system for run-time operational monitoring of an autonomous vehicle*. U.S. Patent (2024).
7. **Nasr, Ali**, Zhu, Kevin, McPhee, John. *Using musculoskeletal models to generate physically-consistent data for 3D human pose, kinematic, dynamic, and muscle estimation*. Journal of

Multibody System Dynamics (2024). [doi:10.1007/s11044-024-10021-5](https://doi.org/10.1007/s11044-024-10021-5)

8. Zhu, Kevin, **Nasr, Ali**, Wong, Alexander, McPhee, John. *3D human pose and torque estimation from monocular video*. (2024).
9. **Nasr, Ali**, McPhee, John. *Inverse kinematic, dynamic, and muscle simulation from video files: Toward markerless 3D human pose, torque, and muscle estimation*. (2024).
10. Korovkina, Vasylysa, **Nasr, Ali**, McPhee, John. *Optimal EMG placement for hand prosthetic control*. (2024).
11. Bell, Sydney, **Nasr, Ali**, McPhee, John. *General muscle torque generator model for a two degree-of-freedom shoulder joint*. Journal of Biomechanical Engineering (2024). [doi:10.1115/1.4065044](https://doi.org/10.1115/1.4065044)
12. **Nasr, Ali**, McPhee, John. *Scalable musculoskeletal model for dynamic simulations of lower body movement*. Journal of Computer Methods in Biomechanics and Biomedical Engineering (2024). [doi:10.1080/10255842.2024.2316240](https://doi.org/10.1080/10255842.2024.2316240)
13. **Nasr, Ali**, Dickerson, Clark R., McPhee, John. *Experimental study of fully passive, fully active, and active-passive upper-limb exoskeleton efficiency: An assessment of lifting tasks*. Journal of Sensors (2023). [doi:10.3390/s24010063](https://doi.org/10.3390/s24010063)
14. **Nasr, Ali**, Bell, Sydney, Whittaker, Rachel L., Dickerson, Clark R., McPhee, John. *Robust machine learning mapping of sEMG signals to future actuator commands in biomechatronic devices*. Journal of Bionic Engineering (2023). [doi:10.1007/s42235-023-00453-8](https://doi.org/10.1007/s42235-023-00453-8)
15. Haraguchi, Naoto, **Nasr, Ali**, Inkol, Keaton A., Hase, Kazunori, McPhee, John. *Human and passive lower-limb exoskeleton interaction analysis: Computational study with dynamics simulation using nonlinear model predictive control*. IEEE (2023). [doi:10.23919/SICE59929.2023.10354196](https://doi.org/10.23919/SICE59929.2023.10354196)
16. McPhee, John, **Nasr, Ali**. *Multibody system dynamics: A fundamental tool for biomechatronic system design*. (2023).
17. **Nasr, Ali**, Hashemi, Arash, McPhee, John. *Scalable musculoskeletal model for dynamic simulations of upper body movement*. Journal of Computer Methods in Biomechanics and Biomedical Engineering (2023). [doi:10.1080/10255842.2023.2184747](https://doi.org/10.1080/10255842.2023.2184747)
18. **Nasr, Ali**, Hunter, Jason, Dickerson, Clark R., McPhee, John. *Evaluation of a machine-learning-driven active-passive upper-limb exoskeleton robot: Experimental human-in-the-loop study*. Journal of Wearable Technologies (2023). [doi:10.1017/wtc.2023.9](https://doi.org/10.1017/wtc.2023.9)
19. **Nasr, Ali**, McPhee, John. *Computational evaluation of active-passive exoskeleton controllers with a scalable biomechatronics model*. (2023).
20. **Nasr, Ali**. *Design, Dynamics, and Control of Active-passive Upper-limb Exoskeleton Robots*. University of Waterloo (2022).
21. **Nasr, Ali**, Bell, Sydney, McPhee, John. *Optimal design of active-passive shoulder exoskeletons: A computational modeling of human-robot interaction*. Journal of Multibody System Dynamics (2022). [doi:10.1007/s11044-022-09855-8](https://doi.org/10.1007/s11044-022-09855-8)
22. Febrer-Nafria, Miriam, **Nasr, Ali**, Ezati, Mahdokht, Brown, Peter, Font-Llagunes, Josep M, McPhee, John. *Predictive multibody dynamic simulation of human neuromusculoskeletal systems: A review*. Journal of Multibody System Dynamics (2022). [doi:10.1007/s11044-022-09852-x](https://doi.org/10.1007/s11044-022-09852-x)
23. **Nasr, Ali**, McPhee, John. *Multibody constrained dynamic modelling of human-exoskeleton: Toward optimal design and control of an active-passive wearable robot*. (2022).
24. **Nasr, Ali**, McPhee, John. *Biarticular MuscleNET: A machine learning model of biarticular muscles*. (2022).
25. **Nasr, Ali**, Ferguson, Spencer, McPhee, John. *Model-based design and optimization of passive shoulder exoskeletons*. Journal of Computational and Nonlinear Dynamics (2022). [doi:10.1115/1.4053405](https://doi.org/10.1115/1.4053405)
26. **Nasr, Ali**, Hashemi, Arash, McPhee, John. *Model-based mid-level regulation for assist-as-needed hierarchical control of wearable robots: A computational study of human-robot adaptation*. Journal of Robotics (2022). [doi:10.3390/robotics11010020](https://doi.org/10.3390/robotics11010020)
27. **Nasr, Ali**, Inkol, Keaton A., Bell, Sydney, McPhee, John. *InverseMuscleNET: alternative machine learning solution to static optimization and inverse muscle modeling*. Journal of Frontiers in Computational Neuroscience (2021). [doi:10.3389/fncom.2021.759489](https://doi.org/10.3389/fncom.2021.759489)
28. **Nasr, Ali**, Bell, Sydney, He, Jiayuan, Whittaker, Rachel L., Dickerson, Clark R., Jiang, Ning, McPhee, John. *MuscleNET: Mapping electromyography to kinematic and dynamic biomechanical*

- variables by machine learning. *Journal of Neural Engineering* (2021). [doi:10.1088/1741-2552/ac1adc](https://doi.org/10.1088/1741-2552/ac1adc)
29. **Nasr, Ali**, He, Jiayuan, Jiang, Ning, McPhee, John. *Muscle modelling using machine learning and optimal filtering of sEMG signals*. (2021).
 30. **Nasr, Ali**, Ferguson, Spencer, McPhee, John. *Model-based design and optimization of passive shoulder exoskeletons*. ASME (2021). [doi:10.1115/DETC2021-69437](https://doi.org/10.1115/DETC2021-69437)
 31. **Nasr, Ali**, Laschowski, Brokoslaw, McPhee, John. *Myoelectric control of robotic leg prostheses and exoskeletons: A review*. ASME (2021). [doi:10.1115/DETC2021-69203](https://doi.org/10.1115/DETC2021-69203)
 32. **Nasr, Ali**, He, Jiayuan, Jiang, Ning, McPhee, John. *Activation torque estimation of muscles by forward neural networks (Forward-MuscleNET) for sEMG-based control of assistive robots*. (2020).
 33. **Nasr, Ali**, McPhee, John. *Control-oriented muscle torque (COMT) model for EMG-based control of assistive robots*. (2020).
 34. **Nasr, Ali**, Arami, Arash, McPhee, John. *Optimal cost function for predicting upper-limb movement with external load*. (2019).
 35. Ghanbari, Masoud, Mousavi, Mohammad Reza, Moosavian, S Ali A, **Nasr, Ali**, Zarafshan, Payam. *Experimental analysis of an optimal redundancy resolution scheme in a cable-driven parallel robot*. IEEE (2017). [doi:10.1109/ICRoM.2017.8466229](https://doi.org/10.1109/ICRoM.2017.8466229)
 36. Mousavi, Mohammad Reza, Ghanbari, Masoud, **Nasr, Ali**, Moosavian, S Ali A, Zarafshan, Payam. *Sensory feedback performance improvement on robocab: An experimental approach to wire-driven parallel manipulator*. IEEE (2016). [doi:10.1109/ICRoM.2016.7886787](https://doi.org/10.1109/ICRoM.2016.7886787)
 37. Nabipour, Maryam Sadat, Arteghzadeh, Nima, Moosavian, S Ali A, **Nasr, Ali**. *Visual servoing in a cable robot using Microsoft Kinect v2 sensor*. IEEE (2016). [doi:10.1109/ICRoM.2016.7886803](https://doi.org/10.1109/ICRoM.2016.7886803)
 38. **Nasr, Ali**, Moosavian, S Ali A. *Multi-objective optimization design of spatial cable-driven parallel robot equipped with a serial manipulator*. *Journal of Modares Mechanical Engineering* (2016). [doi:10.1088/1741-2552/ac1adc](https://doi.org/10.1088/1741-2552/ac1adc)
 39. **Nasr, Ali**. *Multiple Impedance Control of Hybrid Parallel-Serial Manipulator for Object Manipulation*. K.N. Toosi University of Technology (2015).
 40. **Nasr, Ali**, Moosavian, S Ali A. *Multi-criteria design of 6-dof fully-constrained cable driven redundant parallel manipulator*. IEEE (2015). [doi:10.1109/ICRoM.2015.7367591](https://doi.org/10.1109/ICRoM.2015.7367591)
 41. Sajadi, Mohammad Reza, **Nasr, Ali**, Moosavian, S Ali A, Zohoor, Hassan. *Mechanical design, fabrication, kinematics and dynamics modeling, multiple impedance control of a wrist rehabilitation robot*. IEEE (2015). [doi:10.1109/ICRoM.2015.7367799](https://doi.org/10.1109/ICRoM.2015.7367799)
 42. Parandian, Yasaman, Arabshahi, Hannaneh Z, **Nasr, Ali**, Moosavian, S Ali A. *Time optimized digital image processing of ball and plate system using artificial neural network*. IEEE (2015). [doi:10.1109/ICRoM.2015.7367775](https://doi.org/10.1109/ICRoM.2015.7367775)
 43. **Nasr, Ali**, Asadi, Farhad, Azami, Kian, Ziaei-Rad, Saeed. *Optimization of trunk angle in biped robots in generating a stable path in up-stair walking*. Iranian Society of Mechanical Engineers (ISME) (2015).
 44. **Nasr, Ali**, Shahriari, Yasser, Sadati, Sayed Hussein. *Optimization in joint space of six-legged walking robots*. Iranian Society of Mechanical Engineers (ISME) (2014).
 45. **Nasr, Ali**. *Optimizing Joint Space Energy of Redundant Tethered Space Robots*. Isfahan University of Technology (2013).

Projects

- **AI-Powered Exam Preparation & Assessment Platform (Full-Stack Distributed SaaS Architecture)** 2026-04-01 – 2026-05-31
 - Architected an AI-powered examination & assessment platform using Next.js 15, React 18, TypeScript, Node.js, Express.js, PostgreSQL, Firebase Auth, Stripe, and Docker.
 - Designed a layered system architecture with SSR/CSR rendering, REST APIs, secure internal service communication, and containerized deployment via Docker Compose.
- **Fuzzy Control of Truck-Trailer Auto Parking in the Presence of Obstacles** 2013-10-01 – 2013-12-01
 - Developed a hierarchical fuzzy logic controller in MATLAB/Simulink for autonomous reverse parking of a truck-trailer system.

- Modeled the nonlinear kinematics of a 4-DOF trailer-truck vehicle and implemented steering control algorithms for reverse maneuvering.
- **Design and Programming of a 3-DoF Robot Controller Circuit Board** 2013-04-01 – 2013-07-01
 - Designed and developed an embedded control system for a 3-DoF robotic manipulator using an ATmega32 AVR microcontroller.
 - Implemented simultaneous position and motion control of two servo motors and one DC motor to execute predefined robotic trajectories.
- **Control System Design of a Nonlinear Inverted Pendulum System** 2013-04-01 – 2013-06-01
 - Developed a mathematical model of a nonlinear inverted pendulum-on-cart system using Newtonian dynamics and derived governing equations of motion.
 - Linearized the nonlinear system around the upright equilibrium point and formulated both transfer function and state-space representations.
- **Hydraulic Circuit Design for Ring Rolling Machine** 2013-04-01 – 2013-06-01
 - Designed hydraulic circuits for a ring rolling machine based on specified force, stroke, pressure, and speed requirements.
 - Performed hydraulic actuator sizing calculations, including cylinder bore and piston rod diameter selection for multiple machine functions.
- **LabVIEW-Based Simultaneous Control of 3 CNC Stepper Motors** 2013-02-01 – 2013-03-01
 - Designed and developed a LabVIEW-based motion control system for simultaneous operation of 3 stepper motors used in CNC applications.
 - Implemented real-time motor speed control by converting user-defined rotational speed inputs into synchronized timing and pulse-delay sequences.
- **Stable Path Planning and Body Angle Analysis of a Biped Robot** 2012-03-01 – 2012-06-01
 - Analyzed the influence of torso/body angle parameters on the dynamic stability of a biped robot using Zero Moment Point (ZMP) stability criteria.
 - Developed kinematic and dynamic models of a multi-link biped robot, including body segments, joints, mass distribution, and inertia properties.